Hussein El Amouri

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Professional Experience

2025–present **Assistant Professor and Researcher**, *University of Science and Arts- USAL, Lebanon*

Courses: Advanced Data Structure, Data Structure, Introduction to Programming using Java, Network. **Research:** Proposing a new diffusion-based self-supervised large model for seizure prediction with minimal labeling hrough denoising diffusion pre-training and reinforcement learning (RL)-based fine-tuning.

2023–2025 **Assistant Research**, *Icube Laboratory, University of Strasbourg, France*

Examining various approaches to advance incremental constrained clustering within the time series and remote sensing domain, employing active learning methodologies. Leveraging the contributions derived from the PhD contribution to inform and optimize the active process.

2020–2023 **Ph.D. Student**, *Icube Laboratory, University of Strasbourg, France*

Learning interpretable-explainable time series transformation that integrates expert knowledge to facilitate clustering, constrained clustering, and interactive clustering processes. This approach leverages deep learning techniques, including contrastive learning and Siamese networks. Different strategies are provided to explain the transformation and clustering results based on shapelets.

2020–2024 **Lecturer**, *University of Strasbourg, France*

"Culture pour l'informatique," providing essential technical skills to navigate the Linux system, covering simple commands to complex scripts.

“Anglais pour l'informatique" is tailored for third-year license students, offering insights into the research field in computer science. The course emphasizes the handling of English-written papers and focuses on understanding state-of-the-art contributions.

Coaching for a hackathon on AI at Strasbourg University.

2020 **Speech synthesis using AI methods Internship**, *GIPSA-LAB Grenoble, France*
Introducing a methodology to benchmark neural vocoders beyond their conventional training conditions, where we control the ranges of seen/unseen training data. We used four types of vocoder (autoregressive, source filter, flows, GAN).

2019 **Telecommunication Engineering Internship**, *Ogero Center, Beirut, Lebanon*
Hands-on and introduction to Lebanese Telecommunication system, Architecture, and Protocols

Published Articles

2025 **Toward EEG-Free Seizure Detection: Evidence of EEG-ECG Synchronization**
IEEE/ACS 22nd International Conference on Computer Systems and Applications, **AICCSA**, DOI: 10.1109/aiccsa66935.2025.11315411, PDF

2023 **Constrained DTW preserving shapelets for explainable time-series clustering**.
Pattern Recognition journal, **Impact-Factor:8**, DOI: 10.1016/j.patcog.2023.109804, PDF

2022 **Constrained DTW Preserving Shapelets**
ECML conference, **Core-Ranking:A**, DOI: 10.1007/978-3-031-26387-3_2, PDF

2020 **Evaluating the Extrapolation Capabilities of Neural Vocoders to Extreme Pitch Values**
Interspeech conference, **Core-Ranking:A**, DOI: 10.21437/Interspeech.2021-1547, PDF

Education

2020–2023 **Ph.D. candidate in Computer Science**, Icube Laboratory, University of Strasbourg.

2019–2020 **Master in Singal and Image processing (Dual Degree)**, Grenoble INP-Phelma, Grenoble, France.

2015–2020 **Electrical and Telecommunication Engineering**, Lebanese University Faculty , Beirut, Lebanon.

Keywords: Active; Incremental; contrastive Learning, Remote sensing (time series and imaging), Clustering,

References available upon request

Skills (100+ Hours)

Programming:	*Python* | *Bash* | *C/C++* | *Matlab* | *MySQL* | *Assembly*
Libraries:	*Pytorch* | *TensorFlow* | *Numpy* | *Pandas* | *Scikit − Learn* | *Matplotlib*
Technologies:	*Git* | *Slurm* | *Docker* | *Networking utils* | *Cisco networking* certified (CCNA 1-4)
Workshops:	*Scientific communication* | *Research integrity* | *Generative AI*
Languages:	*English* (Fluent) | *Arabic* (Native) | *French* (Intermediate)

Keywords: Active; Incremental; contrastive Learning, Remote sensing (time series and imaging), Clustering,

References available upon request